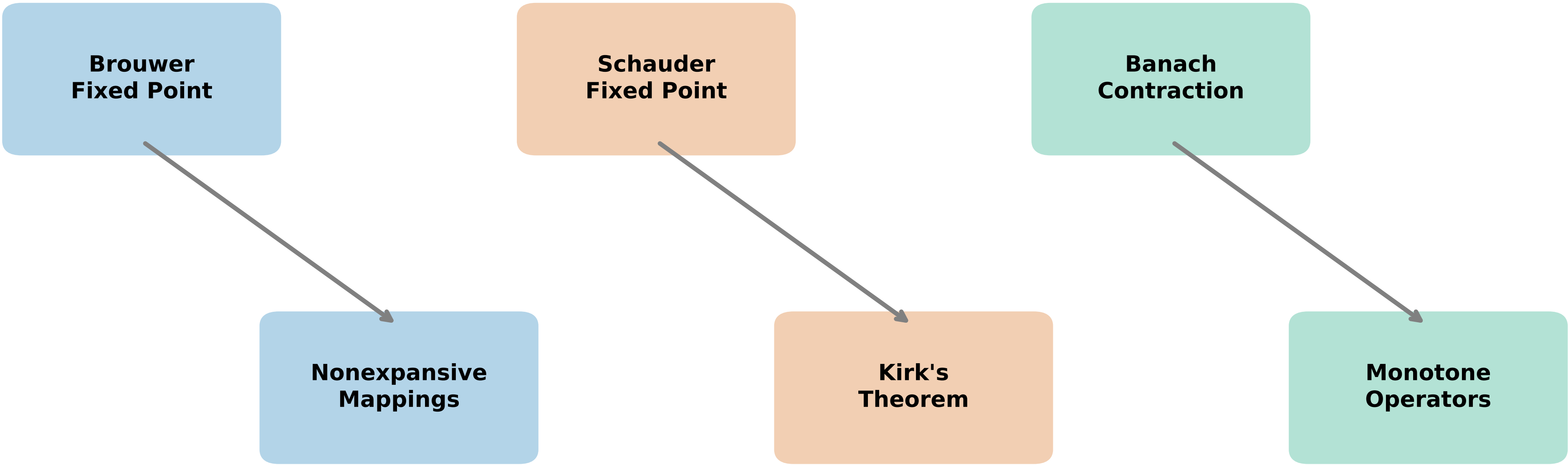
# Hierarchy of Fixed Point Theorems




**Required Conditions for Existence:**

- Brouwer: Compact convex set in $\mathbb{R}^n$
- Schauder: Compact convex set in Banach space
- Contraction: Complete metric space + contraction property
- Nonexpansive: Bounded convex set with normal structure
- Kirk: Uniformly convex Banach space
- Monotone: Monotone + coercivity conditions